# Hands-On Migration with AI Assistance

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# 1. AI-Assisted Migration: Usage and Disclaimer

**Sharing of capabilities, not a formal delivery commitment**

## Partner Training| Huawei Cloud MaaS for VM to ECS migration

**Central Note**

This session presents the AI-assisted capabilities of Huawei Cloud MaaS and their application ideas for migrating VMs from cloud to ECS. It does not process real customer data nor constitute a formal delivery plan, security or compliance commitment, or product feature promise.

### 1. Data and Privacy

**Security / Compliance / Protection**

- No real customer data is collected or processed
- Use simulated, test, or anonymized data
- Do not upload accounts, passwords, keys, certificates, or data

### 2. Usage Limits

**Reference / Human Review**

- AI supports planning, analysis, scripts, and risk assessment
- It does not replace design, expert review, testing, or decisions
- Review plans, scripts, and configurations before the project

### 3. Customer Authorization

**Authorization / Approval**

- The customer decides whether to use AI in real projects
- Without authorization, do not enter customer information into AI
- Respect security, compliance, confidentiality, and approvals

### 4. Not Recommended Scenarios

**Prudent Evaluation / Manual Process**

- Evaluate or avoid highly sensitive security scenarios
- Classified systems, financial core, defense, or energy
- Use manual migration under strict sovereignty or AI prohibition

**AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Real implementation must be based on customer confirmation, security, compliance, and project situation.**

# 2. The Cloud Migration Challenge



**The Current Migration Landscape**

Migrations between public clouds present significant technical and operational complexities. The diversity of services and configurations requires thorough analysis prior to migration. This traditional process consumes time and valuable resources.



**The AI Opportunity**

Artificial intelligence revolutionizes the cloud migration approach through automated analysis. It enables intelligent identification of patterns, dependencies, and requirements. It transforms complex migrations into efficient and predictable processes.



**Workshop Objective**

Demonstrate how AI can reduce migration times from weeks to days. Show real cases of automated source infrastructure analysis. Present the practical methodology for implementing intelligent migrations.



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# 3. How AI Works in Migrations

## Automated Infrastructure Analysis



AI automatically examines all services in the source cloud. It identifies configurations, dependencies, and relationships between services. It generates a complete and detailed inventory without manual intervention.

## Intelligent Service Mapping



Specialized algorithms correlate services across public clouds. They suggest optimal equivalents in Huawei Cloud considering functionality and cost. They learn from previous migrations to improve recommendations.

## Automated Migration Planning



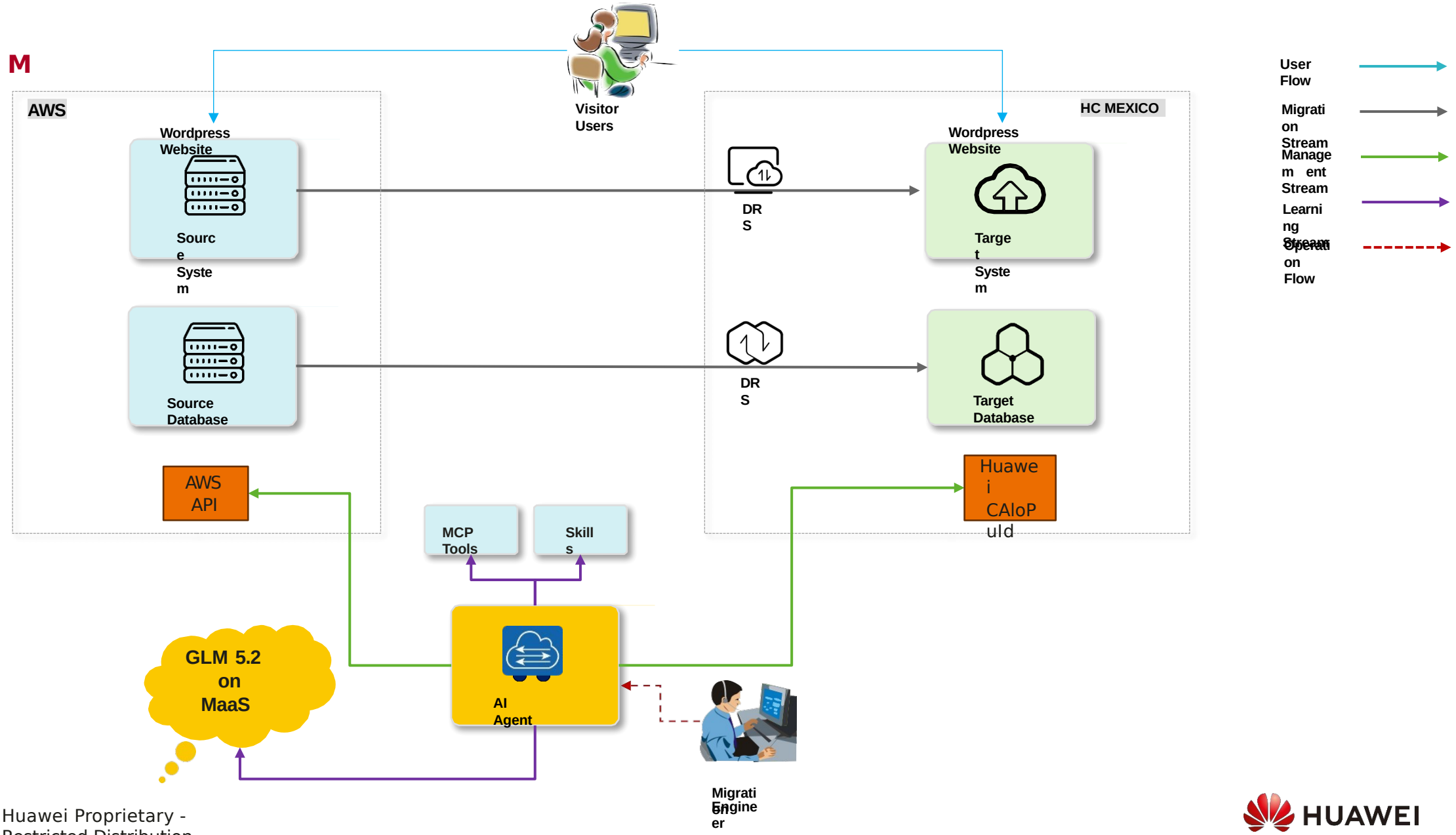
It generates sequential migration plans based on dependencies. It calculates maintenance windows and accurate time estimates. It identifies potential risks and suggests mitigation strategies.



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# 4. DEMO






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## 4.1 AWS to Huawei Cloud Service Mapping

| AWS Service | Huawei Cloud | Tool | Notes |
|---|---|---|---|
| EC2 (m7i-flex.large) | ECS (c6.large.4) | SMS | Block-level server migration |
| RDS MySQL 8.0 (db.t3.micro) | RDS MySQL 8.0 (n1.large.2) | DRS | Full + incremental DB sync |
| S3 (versioning) | OBS | OMS | Object storage migration |
| VPC (10.0.0.0/16) | VPC (10.0.0.0/16) | Terraform | Recreated with IaC |
| Subnets (public/private) | VPC Subnet | Terraform | Recreated with IaC |
| Security Groups | Security Groups | Terraform | HTTP/SSH/MySQL rules |
| Internet Gateway | VPC Gateway | Terraform | Implicit in VPC |
| Route Tables | Route Tables | Terraform | Implicit in VPC |
| Key Pairs (ED25519) | KPS | Terraform | SSH key management |
| EBS (gp3) | EVS (SAS) | SMS | Automatic with server migration |
| CloudWatch | CES | Manual | Post-migration setup |

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# 4.2 Huawei Cloud Migration Tools

## SMS

Server Migration Service
Migrates entire VMs from AWS/Azure to Huawei ECS
Features:
- Block-level disk copy
- Linux & Windows support
- Automatic network config

Demo usage:
- Target server template
- AWS EC2 -> Huawei ECS
- Flavor: c6.large.4
- Volume: SAS type

## DRS

Data Replication Service
Migrates databases between heterogeneous DBMS
Features:
- Full + incremental sync
- MySQL-to-MySQL support
- Online migration

Demo usage:
- DEH connectivity
- AWS RDS MySQL 8.0
- -> Huawei RDS MySQL 8.0
- FULL_INCR_TRANS mode

## OMS

Object Storage Migration
Migrates objects from AWS S3 to Huawei OBS
Features:
- Metadata migration
- Consistency check
- Failed object recording
- Prefix-based filtering

Demo usage:
- AWS S3 bucket
- -> Huawei OBS bucket
- Size+timestamp verify

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## 4.3 Infrastructure as Code: Terraform

### Source: AWS (Terraform)

Provider: hashicors/aws | Region: us-

east-2 Files:
- main.tf - Provider, key pairs, AMI data
- vpc.tf - VPC, subnets, IGW, route tables
- ec2.tf - EC2 instance (WordPress + Apache)
- rds.tf - RDS MySQL 8.0, subnet group
- s3.tf - S3 bucket with versioning
- security_groups.tf - SG for EC2 and RDS

Instance: m7i-flex.large (Ubuntu 22.04)
Database: db.t3.micro, MySQL 8.0, 20GB gp3 WordPress installed via user_data script

### Target: Huawei Cloud (Terraform)

Provider: huaweicloud/huaweicloud v1.93.0 Region: la-north-2 (Mexico)

Modules:
- sms/network/ - VPC, subnet, security groups
- sms/migration/ - SMS template + migration task
- drs/ - RDS instance + DRS job
- oms/ - OBS bucket + OMS migration

ECS: c6.large.4 (via SMS)
RDS: rds.mysql.n1.large.2, MySQL 8.0, 20GB OBS: private bucket (via OMS from S3)
DRS: mysql-to-mysql, FULL_INCR_TRANS

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## 4.4 AI Agent Architecture: GLM 5.2 + MCP

### Migration Stream

AI analyzes source infrastructure
Generates Terraform for target
Executes migration via
SMS/DRS/OMS Monitors progress
and handles errors

### Management Stream

Migration Engineer interacts
with AI Reviews and approves
plans
Adjusts configurations as
needed Controls execution
flow

### Learning Stream

AI learns from each migration
Improves service mapping
accuracy Builds knowledge
base of patterns Reduces future
migration time

### AI Agent: GLM 5.2 on MaaS + MCP Tools

Model: GLM 5.2 running on Huawei Cloud Model as a Service (MaaS)
MCP Tools: AWS API (source inventory) + Huawei Cloud API (target provisioning) + DRS + SMS + OMS
Skills: Migration Engineer knowledge base with cloud service mappings, Terraform templates,
best practices Flow: User request -> AI analyzes source -> Generates plan -> Engineer reviews
-> AI executes -> Verification

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## 5. BENEFITS



### Operational Efficiency

**Reduction of time in analysis and planning according to the source infrastructure.**
**Elimination of human errors in service mapping.**
**Automation of repetitive migration**



### Cost Optimization

**Can generate cost optimization recommendations.**
**Reduction of operational migration costs**



### Risk Reduction

Early detection of risks, incompatibilities, and critical dependencies.



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